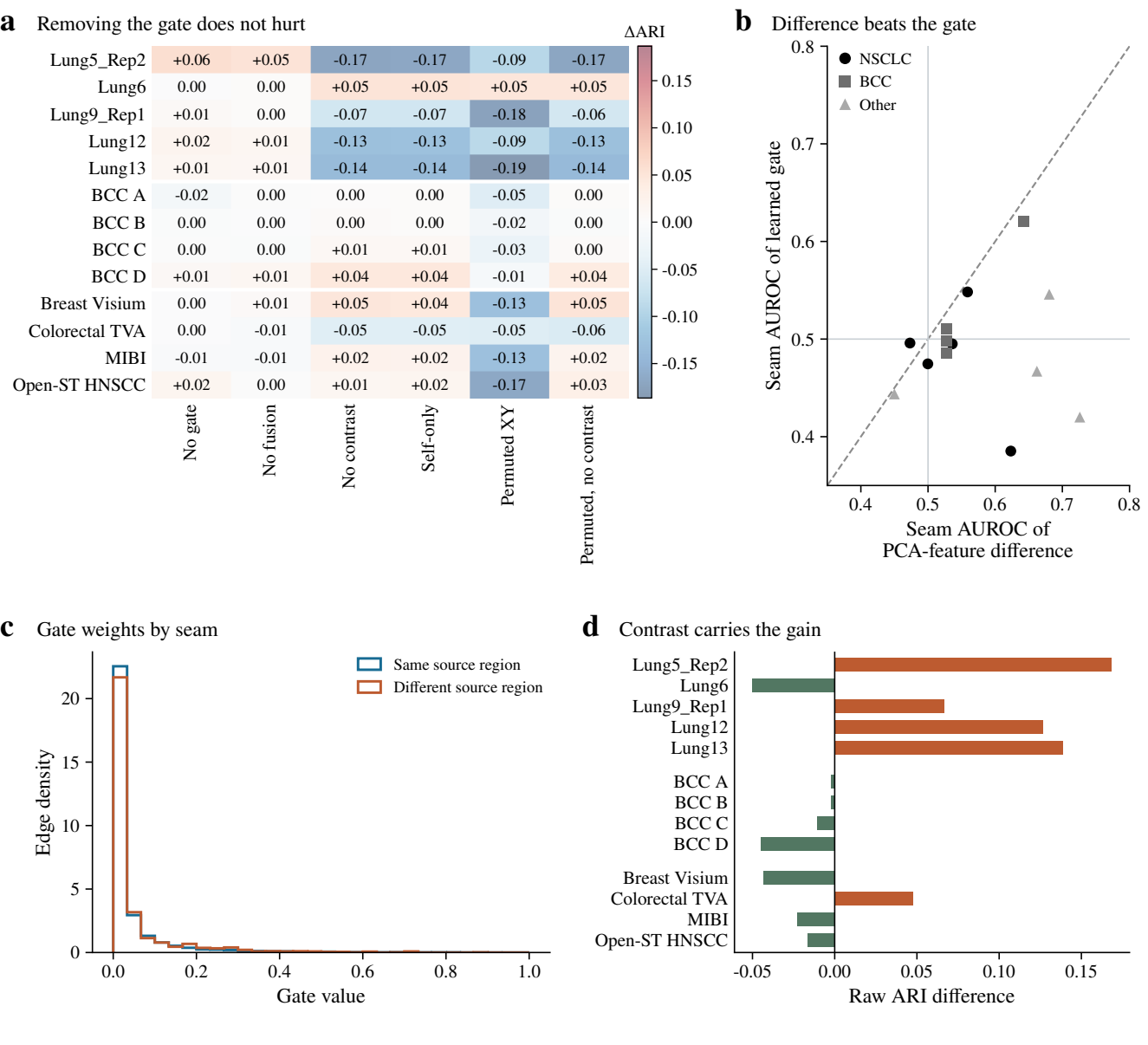

a
Removing the gate does not hurt
ΔARI
Lung5_Rep2 +0.06 +0.05 -0.17 -0.17 -0.09 -0.17
Lung6 0.00 0.00 +0.05 +0.05 +0.05 +0.05
Lung9_Rep1 +0.01 0.00 -0.07 -0.07 -0.18 -0.06
Lung12 +0.02 +0.01 -0.13 -0.13 -0.09 -0.13
Lung13 +0.01 +0.01 -0.14 -0.14 -0.19 -0.14
BCC A -0.02 0.00 0.00 0.00 -0.05 0.00
BCC B 0.00 0.00 0.00 0.00 -0.02 0.00
BCC C 0.00 0.00 +0.01 +0.01 -0.03 0.00
BCC D +0.01 +0.01 +0.04 +0.04 -0.01 +0.04
Breast Visium 0.00 +0.01 +0.05 +0.04 -0.13 +0.05
Colorectal TVA 0.00 -0.01 -0.05 -0.05 -0.05 -0.06
MIBI -0.01 -0.01 +0.02 +0.02 -0.13 +0.02
Open-ST HNSCC +0.02 0.00 +0.01 +0.02 -0.17 +0.03
No gate
No fusion
No contrast
Self-only
Permuted XY
Permuted, no contrast
0.15
0.10
0.05
0.00
-0.05
-0.10
-0.15
b
Difference beats the gate
NSCLC
BCC
Other
Seam AUROC of learned gate
Seam AUROC of PCA-feature difference
0.4
0.5
0.6
0.7
0.8
c
Gate weights by seam
Same source region
Different source region
Edge density
Gate value
0
5
10
15
20
0.0
0.2
0.4
0.6
0.8
1.0
d
Contrast carries the gain
Lung5_Rep2
Lung6
Lung9_Rep1
Lung12
Lung13
BCC A
BCC B
BCC C
BCC D
Breast Visium
Colorectal TVA
MIBI
Open-ST HNSCC
-0.05
0.00
0.05
0.10
0.15
Raw ARI difference